

Revised Generation Interconnection System Impact Study Report

For Queue Project Q24-2387

Larder 100 kV
April 17, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2387. The Interconnection Customer proposes a 58.0 MW natural gas combined cycle facility interconnecting at the Larder 100 kV substation in the Ohio area. The request was received on March 01, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 487 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,473,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Geisel Monument Substation / Beaverley 100 KV	Eastown Rd 230-1 kV Transformer	113.7	187	10.2
Beaverley / Southwest Lima 100 KV line	Eastown Rd 230/1kV TX	113.5	199	10.2
Lenkurt / Southwest Lima 100 KV	Eastown Road 230/1 kV TX	113.3	210	10.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beaverley / Geisel Monument 100 KV L	Rebuild line with higher-capacity conductor	\$2,328,000	55% (\$1,270,000)
Beaverley / Southwest Lima 100 KV line	Rebuild line with higher-capacity conductor	\$3,361,000	55% (\$1,833,000)
Lenkurt / Southwest Lima 100 KV	Rebuild line with higher-capacity conductor	\$6,177,000	55% (\$3,370,000)
	Total allocated to project		\$6,473,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.